

Notes on Threshold Cryptography

A companion technical notebook

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1 Paillier Encryption

Paillier encryption [Pai99] is an additively homomorphic public-key scheme based on the *decisional composite residuosity* (DCR) assumption. Its design contrasts with textbook RSA: multiplying RSA ciphertexts multiplies plaintext representatives, whereas multiplying Paillier ciphertexts adds their exponents and therefore adds plaintexts modulo n .

1.1 Setup and Key Generation

The key generation algorithm proceeds as follows.

1. Sample large primes p, q satisfying $\gcd(pq, (p-1)(q-1)) = 1$, and set $n = pq$.
2. Compute $\lambda \leftarrow \text{lcm}(p-1, q-1)$.
3. Sample $g \leftarrow \mathbb{Z}_{n^2}^*$ uniformly at random.
4. Compute $\mu \leftarrow (L(g^\lambda \bmod n^2))^{-1} \bmod n$, where $L(x) := \lfloor (x-1)/n \rfloor$. (This requires $L(g^\lambda \bmod n^2)$ to be invertible mod n ; resample g if not.)
5. Return $pk \leftarrow (n, g)$ and $sk \leftarrow (\lambda, \mu)$.

Remark 1 (Simplified generator). *The common choice $g = 1 + n$ is always valid and simplifies both key generation and decryption. Since $(1+n)^m \equiv 1 + mn \pmod{n^2}$ (Lemma 1 below), one obtains $L(g^\lambda \bmod n^2) = \lambda \bmod n$, so $\mu = \lambda^{-1} \bmod n$ is computable directly without sampling. The security level is identical; see Remark 2.*

Encryption. $\text{Enc}(pk, m \in \mathbb{Z}_n)$: sample $r \leftarrow \mathbb{Z}_n^*$ and return

$$c \equiv_{n^2} g^m \cdot r^n.$$

Decryption. $\text{Dec}(sk, c)$: compute

$$m \leftarrow L(c^\lambda \bmod n^2) \cdot \mu \bmod n.$$

Correctness. We use the following key lemma.

Lemma 1 (Binomial structure of $\langle 1+n \rangle$). *For any $w \in \mathbb{Z}_{n^2}^*$, write its unique factorization in the direct product $\mathbb{Z}_{n^2}^* \cong \langle 1+n \rangle \times H$ (where $H = \{r^n : r \in \mathbb{Z}_n^*\}$ is the subgroup of n -th powers) as $w = (1+n)^{[w]} \cdot h$ for $[w] \in \mathbb{Z}_n$ and $h \in H$. Then:*

1. $(1+n)^k \equiv 1 + kn \pmod{n^2}$ for all $k \in \mathbb{Z}$.
2. $h^\lambda \equiv 1 \pmod{n^2}$ for all $h \in H$.
3. $L(w^\lambda \bmod n^2) = \lambda[w] \bmod n$.

Proof. (i) By the binomial theorem, $(1+n)^k = \sum_{j=0}^k \binom{k}{j} n^j \equiv 1 + kn \pmod{n^2}$, since all terms with $j \geq 2$ are divisible by n^2 .

(ii) Any $h \in H$ has the form $h = r^n$ for $r \in \mathbb{Z}_n^*$. The element $r^n \in \mathbb{Z}_{n^2}^*$ has order dividing $\phi(n)$ (by Carmichael's theorem applied to n^2 together with the condition $\gcd(n, \phi(n)) = 1$), and $\lambda \mid \phi(n)$, so $h^\lambda = r^{n\lambda} \equiv 1 \pmod{n^2}$.

(iii) Write $w = (1+n)^{[w]} \cdot h$. Then $w^\lambda = (1+n)^{[w]\lambda} \cdot h^\lambda = (1+n)^{[w]\lambda}$ by (ii). Applying (i): $(1+n)^{[w]\lambda} \equiv 1 + [w]\lambda n \pmod{n^2}$, so $L(w^\lambda) = [w]\lambda \bmod n$. $\square$

Correctness of decryption now follows immediately: $c = g^m r^n$ gives $[c] = m[g]$ (since $[r^n] = 0$), so

$$L(c^\lambda) \cdot \mu = \lambda[c] \cdot \mu = \lambda m[g] \cdot (L(g^\lambda))^{-1} = \lambda m[g] \cdot (\lambda[g])^{-1} = m \bmod n.$$

1.2 The DCR Assumption

DCR assumption. Let $n = pq$ be generated as above. For every probabilistic polynomial-time adversary $\mathcal{A}$,

$$\left| \Pr_{r \leftarrow \mathbb{Z}_n^*} [\mathcal{A}(n, r^n \bmod n^2) = 1] - \Pr_{z \leftarrow \mathbb{Z}_{n^2}^*} [\mathcal{A}(n, z) = 1] \right|$$

is negligible in the security parameter.

The n -th powers in $\mathbb{Z}_{n^2}^*$ form the subgroup $H = \{r^n \bmod n^2 : r \in \mathbb{Z}_n^*\}$ of index n and order $\phi(n)$; the DCR assumption says the n cosets of H are computationally hidden.

For $g = 1 + n$, IND-CPA security follows by a direct hybrid. Given a DCR challenge z , a reduction chooses a random bit b and gives the adversary

$$c^* = (1+n)^{m_b} z \bmod n^2.$$

If $z = r^n$, this is a correctly distributed encryption of m_b . If z is uniform in $\mathbb{Z}_{n^2}^*$, multiplication by the fixed element $(1+n)^{m_b}$ preserves uniformity, so c^* is independent of b . Any non-negligible IND-CPA advantage therefore distinguishes the two DCR distributions. This is also the usual equivalence between semantic security and IND-CPA for public-key encryption, specialized to Paillier's ciphertext distribution.

Remark 2 (Security with $g = 1 + n$). *Using $g = 1 + n$ does not weaken security. The ciphertext $c = (1 + mn) \cdot r^n$ is uniformly random within the coset $(1 + mn) \cdot H$. Under DCR, these cosets are computationally indistinguishable, so an adversary cannot determine m from c regardless of which valid g is used. The coset structure is determined entirely by r^n , not by the choice of g .*

1.3 Additive Homomorphism

For ciphertexts $c_1 = \text{Enc}(m_1, r_1)$ and $c_2 = \text{Enc}(m_2, r_2)$:

$$c_1 \cdot c_2 \bmod n^2 = g^{m_1} r_1^n \cdot g^{m_2} r_2^n = g^{m_1+m_2} (r_1 r_2)^n = \text{Enc}(m_1 + m_2 \bmod n, r_1 r_2).$$

Scalar multiplication follows similarly:

$$c_1^k \bmod n^2 = g^{km_1} (r_1^k)^n = \text{Enc}(k \cdot m_1 \bmod n, r_1^k).$$

Both operations require only the public key and leave the randomness well-distributed.

2 Threshold Paillier: Construction Sketch

Threshold Paillier distributes decryption authority among ℓ parties so that a qualified set can jointly decrypt, while a coalition below the threshold cannot decrypt on its own. The presentation below isolates the reconstruction algebra used by standard threshold variants; complete schemes also specify modulus generation, verifiable shares, and a security proof [FPS00, DJ01].

2.1 Motivation

Threshold decryption removes a single decryption authority. It is useful when encrypted data are aggregated publicly but opening the result must require agreement, as in electronic voting, sealed-bid auctions, custody systems, and multiparty computation. Homomorphic evaluation can proceed under the public key, while policy is enforced only at decryption time.

2.2 Key Generation and Share Distribution

We describe the *trusted dealer* model; see Remark 4 for the DKG variant.

1. Run Paillier key generation to obtain $pk = (n, g)$ and $sk = (\lambda, \mu)$. Publish pk .
2. Sample a random degree- $(t-1)$ polynomial $f \in \mathbb{Z}_{\phi(n)n}[X]$ with $f(0) = \phi(n)$ (so $sk = \phi(n)$).
3. For each party $i \in [\ell]$, set $sk_i \leftarrow f(i) \bmod \phi(n)n$ and send sk_i privately to party i .
4. Erase f and sk .

Remark 3 (Composite-ring sharing). *This interpolation takes place over the composite ring $\mathbb{Z}_{\phi(n)n}$, not a field. It is therefore not ordinary Shamir sharing, and perfect privacy cannot be inferred from the standard field proof. Threshold Paillier constructions choose the sharing modulus and secret congruences together so that the required integer interpolation is well defined, and establish privacy by reduction to the underlying encryption assumption. The formulas here should be read as the reconstruction layer of that construction, not as a stand-alone generic secret-sharing scheme.*

Set $\Delta := \ell!$. For any subset $S \subseteq [\ell]$ with $|S| = t$ and any $i \in S$, define the *normalized Lagrange coefficient*

$$\Lambda_i(0) := \Delta \cdot \lambda_i^S(0) = \Delta \prod_{j \in S \setminus \{i\}} \frac{-j}{i-j} \in \mathbb{Z},$$

which is always an integer since all denominators divide $\Delta = \ell!$.

Remark 4 (DKG variant). *In a distributed-key-generation variant, the parties jointly generate the RSA modulus and compatible decryption shares without reconstructing p , q , or the complete secret key at any one party. This is a separate protocol problem and is substantially more involved than distributing shares from a trusted dealer. The construction sketch in this chapter assumes the dealer model.*

2.3 Partial Decryption and Combining

Given a ciphertext c and a set S of t participating parties, decryption proceeds as follows.

Partial decryption. Each party $i \in S$ computes its *decryption share*:

$$c_i \equiv_{n^2} c^{2\Delta \cdot sk_i}.$$

(The factor of 2 ensures that the relevant values lie in the quadratic-residue subgroup used by the threshold construction [DJ01].)

Combining. The combiner computes:

$$\hat{m} \equiv_{n^2} \prod_{i \in S} c_i^{2\Lambda_i(0)}.$$

Since $\sum_{i \in S} sk_i \cdot \Lambda_i(0) = \Delta \cdot sk$ by Lagrange interpolation, the product evaluates to $c^{4\Delta^2 \cdot sk}$. Applying L and scaling by the appropriate precomputed constant recovers m .

Why integer exponents suffice. The Lagrange coefficients $\lambda_i^S(0)$ are rational in general. Multiplying by $\Delta = \ell!$ clears all denominators (since differences $i - j$ for $i, j \in [\ell]$ divide $\ell!$), making $\Lambda_i(0) \in \mathbb{Z}$. This avoids the need to compute modular inverses in $\mathbb{Z}_{n^2}^*$, which would require knowing $\phi(n^2)$ and hence the factorization of n — precisely the secret we are trying to protect.

2.4 Semi-Honest Security and Simulation

The scheme above is secure against *semi-honest* adversaries who corrupt up to $t - 1$ parties and follow the protocol honestly but try to learn the plaintext from their views. Security is proved by a reduction to plain Paillier IND-CPA security.

The reduction. The reduction is given a plain Paillier public key N and access to a CPA challenge oracle. It must simulate the entire threshold decryption game for an adversary A who corrupts a set $C \subseteq [\ell]$ with $|C| \leq t - 1$.

1. **Setup.** Sample $\Theta \leftarrow \mathbb{Z}_N$ and set $pk = (N, \Theta)$. For each corrupted party $i \in C$, sample $sk_i \leftarrow \mathbb{Z}_{N^2}$ uniformly and hand these to A . The reduction does not know the real $sk = \phi(N)$.

2. **Decryption queries.** When A asks for a decryption of message m , the reduction:

- (a) Samples $r \leftarrow \mathbb{Z}_N$ and computes $c \equiv_{N^2} (1 + mN) \cdot r^N$ (possible using only $pk = N$).
- (b) For each honest party $j \notin C$, computes the simulated share:

$$c_j \equiv_{N^2} (1 + Nm\Theta)^{\Lambda_0(j)} \cdot \prod_{i \in C} c^{sk_i \cdot \Lambda_i(j)},$$

where $\Lambda_0(j) = \Delta \prod_{k \in C} \frac{j-k}{0-k}$ and $\Lambda_i(j) = \Delta \cdot \frac{j}{i} \prod_{k \in C \setminus \{i\}} \frac{j-k}{i-k}$ are normalized Lagrange coefficients for evaluating at j from the set $C \cup \{0\}$.

The key insight is that by Lemma 1, $c^{sk} = (1 + mN)^{\phi(N)} \equiv 1 + mN\phi(N) \equiv 1 + mN\Theta \pmod{N^2}$ (since $\Theta = \phi(N) \pmod{N}$ in the real scheme, and here Θ plays this role). So the simulator expresses the honest party's contribution via m and Θ alone, without needing sk .

3. **Challenge.** A sends $(m_0, m_1) \in \mathbb{Z}_N^2$. The reduction forwards the pair

$$(\Theta\Delta^2 m_0 \bmod N, \Theta\Delta^2 m_1 \bmod N)$$

to the plain Paillier CPA oracle and receives c_b^* . It forwards c_b^* to A as the challenge ciphertext, along with simulated shares computed as above. The messages are scaled by $\Theta\Delta^2$ because the simulation introduces these extra factors: the combiner ultimately recovers $m\Delta^2\Theta \bmod N$ rather than m , so the challenge must be embedded accordingly.

4. **Output.** A outputs a guess b' ; the reduction forwards it to the CPA challenger. Any advantage of A translates directly to an advantage against plain Paillier IND-CPA.

Remark 5 (Why the simulation works). *The simulated shares c_j are indistinguishable from real ones because the honest party's share is determined by Lagrange interpolation — from A 's view (which includes only the corrupted shares and the output), there is exactly one consistent value for each honest share, and the simulator produces exactly that value. No information about sk leaks beyond what is already implied by the corrupted shares and m .*

2.5 Achieving Malicious Security via Verifiable Partial Decryptions

The semi-honest scheme above fails against *malicious* adversaries: a corrupt party can submit an arbitrary $c'_i \neq c^{2\Delta \cdot sk_i}$ as its decryption share, causing the combiner to reconstruct a bogus plaintext with no way to detect the fault. To achieve malicious security, each partial decryption must come with a proof of correctness.

Intuition. At setup the dealer publishes, alongside each share sk_i , a *verification key* $pk_i = v^{\Delta \cdot sk_i}$, where v is a public generator of the subgroup $\mathbb{Q}_{N^2}$ of quadratic residues modulo N^2 . When party i submits $c_i \equiv_{N^2} c^{2\Delta \cdot sk_i}$, any combiner can check consistency by verifying that c_i^2 and pk_i share the same discrete logarithm — namely $\Delta \cdot sk_i$ — with respect to the bases $c^{4\Delta}$ and v^Δ respectively. This is a *proof of discrete-log equality (DLEQ)*: a zero-knowledge argument that

$$\log_{c^{4\Delta}} c_i^2 = \log_{v^\Delta} pk_i = sk_i.$$

A forged share $c'_i \neq c^{2\Delta \cdot sk_i}$ would require producing a valid DLEQ proof for an incorrect relation, which is computationally infeasible under the discrete-logarithm assumption in $\mathbb{Q}_{N^2}$.

Why safe primes and a special g ? Standard DLEQ / Chaum–Pedersen proofs require working in a group of *known* prime order. In $\mathbb{Z}_{N^2}^*$ the group order $\phi(N^2)$ is unknown (knowing it factors N). Two modifications address this:

1. **Safe primes.** Choose $p = 2p' + 1$ and $q = 2q' + 1$ with p', q' prime. Then $\mathbb{Q}_{N^2}$ has order $Np'q' = N\phi(N)/4$. Even though this order is unknown to the prover and verifier, the group has no small subgroups, which is the key property needed for the soundness argument.
2. **Special generator.** Pick $g \equiv_{N^2} (1 + N)^a b^N$ for $b \in \mathbb{Z}_N^*$ and set $v = g^{2\alpha}$ for random $\alpha \in \mathbb{Z}_N^*$. This ensures v generates $\mathbb{Q}_{N^2}$ and, crucially, the reduction knows $\log_g v = 2\alpha$, which is needed to embed the DLEQ proof into a security argument against a plain-Paillier CPA adversary.

Extra factors of 2 and 4. The exponent 2Δ (rather than Δ) in $c_i = c^{2\Delta \cdot sk_i}$ ensures $c_i \in \mathbb{Q}_{N^2}$ (squaring maps any element into the QR subgroup), so that DLEQ can be stated entirely within $\mathbb{Q}_{N^2}$. The DLEQ proof itself introduces a further squaring, so the combiner ultimately recovers $c^{4\Delta^2 \cdot sk}$ and must divide out the extra factor of $4\Delta^2$ to obtain m .

The share proof follows the Chaum–Pedersen equality-of-discrete-log pattern [CP93]: the prover commits using fresh exponent randomness, receives a challenge, and returns a linear response that is checked against both base–target pairs. In an unknown-order group the challenge and response ranges must be bounded explicitly; otherwise the ordinary prime-order special-soundness argument does not apply. The exact range proof and soundness analysis are part of the cited threshold-Paillier constructions rather than the elementary derivation above [FPS00, DJ01].

2.6 Security

In the threshold IND-CPA experiment, an adversary receives the public key and the internal shares of fewer than t corrupted parties, chooses two messages, and receives an encryption of one of them. Subject to the decryption-oracle restrictions of the chosen model, its advantage in guessing the challenge bit must be negligible. Robust variants additionally require that malformed decryption shares be detected and that any qualified set of valid shares reconstruct the same plaintext. These properties depend on the complete sharing, verification, and setup protocols; they do not follow from the interpolation identity alone [FPS00, DJ01].

A Chinese Remainder Theorem and the Structure of $\mathbb{Z}_N^*$

A.1 The Ring Isomorphism

Let $N = pq$ with $\gcd(p, q) = 1$. The **Chinese Remainder Theorem (CRT)** gives a ring isomorphism φ :

$$\mathbb{Z}_N \cong \mathbb{Z}_p \times \mathbb{Z}_q, \quad a \mapsto (a \bmod p, a \bmod q).$$

This is a ring isomorphism: it respects both addition and multiplication. Restricting to units gives a group isomorphism:

$$\mathbb{Z}_N^* \cong \mathbb{Z}_p^* \times \mathbb{Z}_q^*.$$

Proof. Homomorphism. Reduction mod p and mod q each preserve $+$ and $\times$, so φ does too.

Injective. Suppose $\varphi(a) = (0, 0)$, i.e. $p \mid a$ and $q \mid a$. Since $\gcd(p, q) = 1$, we get $pq \mid a$, hence $a \equiv 0 \pmod{N}$. So $\ker \varphi = \{0\}$.

Surjective. Both sides have the same finite size: $|\mathbb{Z}_N| = N = pq = |\mathbb{Z}_p \times \mathbb{Z}_q|$. Injectivity on finite sets of equal size implies surjectivity. $\square$

Remark 6. The only non-trivial step is injectivity, which uses $\gcd(p, q) = 1$. This is why CRT fails for prime powers: $\mathbb{Z}_{p^2} \not\cong \mathbb{Z}_p \times \mathbb{Z}_p$, since $p \mid a$ and $p \mid a$ does not imply $p^2 \mid a$.

A.2 The Reconstruction Formula and Idempotents

The inverse of the CRT map is explicit. Define the *CRT basis elements*:

$$e_p = q \cdot (q^{-1} \bmod p) \bmod N, \quad e_q = p \cdot (p^{-1} \bmod q) \bmod N.$$

These exist because $\gcd(q, p) = \gcd(p, q) = 1$. Given $a_p = a \bmod p$ and $a_q = a \bmod q$, the unique lift to $\mathbb{Z}_N$ is:

$$a = a_p \cdot e_p + a_q \cdot e_q \bmod N. \quad (\text{CRT reconstruction})$$

Key observation: e_p and e_q are idempotents. An element e is idempotent if $e^2 = e$. To verify $e_p^2 = e_p \pmod{N}$, equivalently check $e_p(e_p - 1) \equiv 0 \pmod{N}$:

- **Mod p :** $e_p = q \cdot q^{-1} \equiv 1 \pmod{p}$, so $e_p - 1 \equiv 0 \pmod{p}$. ✓
- **Mod q :** e_p has a factor of q , so $e_p \equiv 0 \pmod{q}$. ✓

Since $p \mid e_p(e_p - 1)$ and $q \mid e_p(e_p - 1)$ and $\gcd(p, q) = 1$, we get $N \mid e_p(e_p - 1)$. Similarly $e_q^2 = e_q$.

Furthermore, $e_p \cdot e_q \equiv 0 \pmod{N}$: the product has a factor of q from e_p and a factor of p from e_q , giving a factor of $pq = N$.

The idempotent decomposition of 1. Note that:

$$e_p + e_q \equiv 1 \pmod{N}.$$

This is because $(e_p + e_q) \equiv 1 \pmod{p}$ and $(e_p + e_q) \equiv 1 \pmod{q}$, so by CRT it equals 1 mod N . Together, $\{e_p, e_q\}$ behave like $(1, 0)$ and $(0, 1)$ in $\mathbb{Z}_p \times \mathbb{Z}_q$ — they are the identity elements of the two components.

Remark 7 (Relation to Shamir secret sharing). *CRT reconstruction and Shamir reconstruction both express a global value as a linear combination of local coordinates. The analogy stops there. CRT coordinates are residues modulo pairwise coprime moduli, and the coefficients are orthogonal idempotents. Shamir shares are evaluations of one low-degree polynomial over a field, and the coefficients are Lagrange interpolation weights. In particular, Shamir's threshold privacy follows from the undetermined coefficients of a random polynomial; it does not follow merely from CRT.*

A.3 Squaring in $\mathbb{Z}_N^*$ via CRT

The idempotent structure makes squaring transparent. If $x = x_p \cdot e_p + x_q \cdot e_q$, then:

$$x^2 = x_p^2 \cdot e_p^2 + 2x_p x_q \cdot e_p e_q + x_q^2 \cdot e_q^2 = x_p^2 \cdot e_p + x_q^2 \cdot e_q \pmod{N},$$

since $e_p^2 = e_p$, $e_q^2 = e_q$, and $e_p e_q = 0$. So squaring in $\mathbb{Z}_N^*$ is just squaring component-wise in $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$.

A.4 Quadratic Residues in $\mathbb{Z}_N^*$

Proposition 1. $a \in \mathbb{Z}_N^*$ is a quadratic residue mod N if and only if it is a quadratic residue mod p and mod q .

Proof. By the squaring observation above: $a = x^2$ in $\mathbb{Z}_N^*$ iff $(a_p, a_q) = (x_p^2, x_q^2)$ in $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$, which holds iff a_p is a QR mod p and a_q is a QR mod q . $\square$

Since exactly $\frac{p-1}{2}$ elements of $\mathbb{Z}_p^*$ are QR mod p (each QR has exactly two square roots $\pm x_p$), the number of QRs in $\mathbb{Z}_N^*$ is:

$$\frac{p-1}{2} \cdot \frac{q-1}{2} = \frac{\phi(N)}{4}.$$

Moreover, each QR mod N has exactly *four* square roots in $\mathbb{Z}_N^*$: the four combinations $(\pm x_p, \pm x_q)$ reconstructed via CRT.

Knowing the factorization $N = pq$ makes residuosity testing and square-root extraction efficient: test each CRT component with the corresponding prime-field criterion, extract roots modulo p and q , and recombine them. Conversely, an algorithm that returns square roots of random quadratic residues modulo a Blum integer yields a probabilistic factoring algorithm: for random x , obtain a root y of x^2 ; whenever $y \not\equiv \pm x \pmod{N}$, the value $\gcd(x - y, N)$ is a nontrivial factor. This is a reduction for square-root extraction, not a blanket claim that every formulation of the quadratic-residuosity decision problem is equivalent to factoring.

A.5 Summary: What CRT Buys You

Object in $\mathbb{Z}_N$	Meaning in $\mathbb{Z}_p \times \mathbb{Z}_q$
e_p, e_q	Primitive idempotents $(1, 0), (0, 1)$
$e_p + e_q = 1$	Partition of unity
$e_p \cdot e_q = 0$	Components are orthogonal
$a^2 \bmod N$	$(a_p^2 \bmod p, a_q^2 \bmod q)$
QR mod N	QR mod p and mod q
# square roots of a QR	$4 = 2 \times 2$

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